

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of M.Pharm (Quality Assurance)

University Theory Examination December 2015

PH816.01 MODERN ANALYTICAL TECHNIQUES

Date: 16.12.2015, Wednesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Explain principle of UV-vis spectroscopy. 05
B) Give principle of IR spectroscopy. What is FT-IR. 05
Q.2 A) Write a Principle and Application of NMR Spectroscopy. 05
B) What is Chemical Shift? Explain Factors affecting Chemical Shift. 05

OR

- Q.2 A) What is coupling constant? Explain its significance in interpretation with suitable examples 05
B) Explain in brief about Derivative UV Spectroscopy with its application. 05
Q.3 Answer the following (Any three) 15
(A) Enlist the ionization techniques used in mass spectroscopy. Discuss MALDI techniques
(B) Define the terms: Mc-Lafferty rearrangement & Metastable peak.
(C) Explain in detail hyphenation technique, LC-MS.
(D) Discuss factors affecting UV-absorption spectra.

SECTION-II

- Q.4 (A) Discuss factors affecting intensity of excitation and emission in spectrofluorimetry. 05
(B) Discuss Principle and application of Differential Scanning calorimetry (DSC). 05
Q.5 (A) Explain how will you determine the degree of crystallinity of a given substance using x-ray method. 05
(B) Write a not on Instrumentation of HPTLC. 05

OR

- Q.5 (A) Explain and derive Bragg's law. 05
(B) Discuss principle of HPLC and its advantages. 05
Q.6 Answer the following (Any three) 15
(A) Explain plate theory for chromatography with column efficiency parameters.
(B) What is enzyme immune-assay? Describe double sandwich ELISA technique for antigen measurement.
(C) Write a note on principle involved in capillary Electrophoresis and Isoelectric focusing.
(D) Write a note on SFC (Super fluid chromatography).